NAU NORTHERN ARIZONA UNIVERSITY
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Memorandum

MA6: Sensors and Actuators

to: course staff

from: project team

context: undergraduate engineering project

Re: MA6: Sensors and Actuators

1.1) What is the minimum gain needed to make the flywheel motor turn on?

The minimum gain needed to make the flywheel motor turn on was 1, but it required a ~20 degree tilt to be noticeable.

1.2)What is the minimum gain needed to start to see improvement in the motorcycle balance?

The minimum gain needed to start seeing improvements was 10.

1.3)Is there an upper limit to the gain at which point the motorcycle balance starts to decrease?

Yes, any gain greater than ~50 saw diminishing returns.

2.1) What is the minimum gain needed to start to see improvement in the motorcycle balance?

The minimum gain needed to start seeing improvements was only 1.

2.2) Is there an upper limit to the gain at which point the motorcycle balance starts to decrease?

Yes, any gain greater than ~5 hurt the motorcycle's balance.

3.1)Which control approach did you implement?

We used the gain Kf, damping the motor's speed by opposing the measured rotation. In testing, we found that Kf provided better results than Ki.

3.2) What are some of the potential problems or concerns with this approach and how did you mitigate them?

A large Kf can oppose the flywheel motion too much, so it was a process of trial-and-error to find the right value.

4. Final System Tuning

We tested each gain level three times and recorded how long the motorcycle balanced, in seconds.

	Kp Gain 1: 1	Kp Gain 2: 10	Kp Gain 3: 50	Kp Gain 4: 75
Trial 1 Time (s)	0.42	1.32	3.70	0.90
Trial 2 Time (s)	0.81	1.06	3.10	1.80
Trial 3 Time (s)	0.67	1.20	2.90	1.10
Average time (s)	0.63	1.19	3.23	1.27

Testing Derivative Control with Kp Gain = 50				
	Kd Gain 1: 1	Kd Gain 2: 10	Kd Gain 3: 5	Kd Gain 4: 7
Trial 1 Time (s)	1.35	1.06	3.41	1.46
Trial 2 Time (s)	1.12	0.95	1.62	0.90
Trial 3 Time (s)	1.42	0.71	1.58	1.01
Average time (s)	1.30	0.91	2.20	1.12

Testing Dampening Control Gain with Kp Gain = 50 and Kd Gain = 5				
	Kf Gain 1: 1	Kf Gain 2: 5	Kf Gain 3: 2.5	Kf Gain 4: 0.5
Trial 1 Time (s)	2.32	0.74	0.51	0.92
Trial 2 Time (s)	2.90	0.71	0.80	0.52
Trial 3 Time (s)	2.27	0.59	0.62	1.3
Average time (s)	2.50	0.68	0.64	0.91

4.1. As you changed the gains, what did you observe?

We observed that very small changes in gain can greatly affect the bike's balancing. We also found that to find the best gain, we usually had to overshoot.

4.2. What combination of gains is most effective for balancing your motorcycle?

The combination that seemed most effective overall was Kp = 50, Kd = 5, and Kf around 1.

Team Contributions

one team member focused on controller software while the other reviewed logic. both contributed to tuning. during trials, one member enabled balance and started a timer while the other held the bike upright and released it. this workflow supported consistent resets between attempts.